

Chapter 1

RTL Design & Implementation of Direct Digital Synthesis Engine

1.1 IP Block 5: Direct Digital Synthesis (DDS) Synthesizer

Generating precise, stable, and frequency-agile analog waveforms is a critical requirement in modern telecommunications and signal processing. Traditional analog oscillators (such as RC circuits or Phase-Locked Loops) suffer from phase-settling delays and are highly susceptible to Process, Voltage, and Temperature (PVT) variations.

To overcome these physical limitations, a **Direct Digital Synthesis (DDS)** engine was developed. This IP block utilizes pure digital logic to mathematically synthesize a continuous analog sine wave, granting the system instantaneous, zero-latency frequency hopping capabilities.

1.1.1 Detailed Working Mechanism & Theory of Operation

The DDS architecture operates on the principle of high-speed phase accumulation, converting a stable reference clock into a highly agile analog waveform through three distinct mathematical stages:

1. The Phase Accumulator (The Core Engine):

At the heart of the DDS is an N -bit synchronous phase accumulator (in this design, $N = 16$). On every active edge of the master clock (f_{clk}), a digital value known as the **Tuning Word** (M) is added to the accumulator's current state.

The accumulator acts as a digital phase wheel. As it continuously adds the tuning word, the 16-bit register eventually exceeds its maximum capacity (65,535) and naturally wraps around to zero. One complete wrap-around corresponds to one full 360° (2π radians) cycle of the output sine wave.

The exact output frequency (f_{out}) is strictly defined by the following mathematical relationship:

$$f_{out} = \frac{M \times f_{clk}}{2^N}$$

A larger Tuning Word (M) causes the accumulator to overflow faster, generating a higher output frequency. A smaller Tuning Word results in slower overflows and a lower frequency. Because this is governed entirely by digital addition, the frequency can be altered instantaneously on a single clock cycle without any analog settling time.

2. Phase Truncation (Area Optimization):

Routing a full 16-bit phase bus directly into a memory lookup table would require 2^{16} (65,536) discrete memory entries, consuming massive and unjustifiable silicon area. To achieve high frequency resolution without the memory overhead, the design employs **Phase Truncation**.

The lower 11 bits are retained internally to maintain mathematical precision over thousands of clock cycles, while only the top 5 bits (`phase_acc[15:11]`) are extracted. This effectively compresses the 65,536 micro-angles into 32 distinct, addressable phase "slices".

3. The Sine Lookup Table (Phase-to-Amplitude Conversion):

The phase accumulator generates a linear digital sawtooth wave. To synthesize a natural sine wave, this linear phase must be translated into non-linear amplitude. The truncated 5-bit phase slice acts as the address index for a combinational Read-Only Memory (ROM) block. The digital multiplexer logic instantly looks up the corresponding 8-bit amplitude (scaled between 0 and 255) for that specific angle, perfectly shaping the final analog curve.

1.1.2 Real-Time Applications

Because DDS is fully digital, it provides unmatched agility in advanced mixed-signal deployments:

- **Software-Defined Radio (SDR) & 5G:** Enables instantaneous RF channel switching and complex digital modulation (PSK, FSK, QAM) by dynamically altering the phase accumulator parameters.
- **FMCW Radar Systems:** Generates mathematically perfect, linear frequency sweeps (chirps) necessary for automotive collision avoidance and targeting systems.
- **Arbitrary Waveform Generators (AWGs):** Serves as the core driving engine for high-end laboratory test equipment, playing back custom waves stored in localized RAM.

1.1.3 Complete Synthesizable Verilog RTL Description

The following synthesizable RTL implements the 16-bit phase accumulator and the optimized 32-sample Sine LUT.

```
'timescale 1ns / 1ps

module dds_sine (
    input wire      clk,           // High-speed system clock
    input wire      rst_n,         // Active-low asynchronous reset
    input wire [15:0] tuning_word, // Controls the output frequency

    output reg [7:0] sine_out      // 8-bit Digital Sine Wave Output
);

//-----
// 1. THE PHASE ACCUMULATOR
//-----
reg [15:0] phase_acc;

always @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        phase_acc <= 16'd0;
    end else begin
        // Continuously add the tuning word.
        // Naturally overflows to create a sawtooth phase.
        phase_acc <= phase_acc + tuning_word;
    end
end

// Extract the top 5 bits of the accumulator to act as our LUT address (0 to 31)
```

```

wire [4:0] lut_index = phase_acc[15:11];

//-----
// 2. THE SINE WAVE LOOKUP TABLE (LUT)
//-----
// Maps the linear phase (0 to 31) into a non-linear sine wave amplitude (0 to 255)

always @(*) begin
    case (lut_index)
        // Quadrant 1 (0 to 90 degrees)
        5'd0  : sine_out = 8'd128;
        5'd1  : sine_out = 8'd153;
        5'd2  : sine_out = 8'd177;
        5'd3  : sine_out = 8'd199;
        5'd4  : sine_out = 8'd218;
        5'd5  : sine_out = 8'd234;
        5'd6  : sine_out = 8'd246;
        5'd7  : sine_out = 8'd253;

        // Quadrant 2 (90 to 180 degrees)
        5'd8  : sine_out = 8'd255; // Peak Positive
        5'd9  : sine_out = 8'd253;
        5'd10 : sine_out = 8'd246;
        5'd11 : sine_out = 8'd234;
        5'd12 : sine_out = 8'd218;
        5'd13 : sine_out = 8'd199;
        5'd14 : sine_out = 8'd177;
        5'd15 : sine_out = 8'd153;

        // Quadrant 3 (180 to 270 degrees)
        5'd16 : sine_out = 8'd128; // Zero Crossing
        5'd17 : sine_out = 8'd103;
        5'd18 : sine_out = 8'd79;
        5'd19 : sine_out = 8'd57;
        5'd20 : sine_out = 8'd38;
        5'd21 : sine_out = 8'd22;
        5'd22 : sine_out = 8'd10;
        5'd23 : sine_out = 8'd3;

        // Quadrant 4 (270 to 360 degrees)
        5'd24 : sine_out = 8'd0; // Peak Negative
        5'd25 : sine_out = 8'd3;
        5'd26 : sine_out = 8'd10;
        5'd27 : sine_out = 8'd22;
        5'd28 : sine_out = 8'd38;
        5'd29 : sine_out = 8'd57;
        5'd30 : sine_out = 8'd79;
        5'd31 : sine_out = 8'd103;

        default : sine_out = 8'd128; // Safe fallback
    endcase
end

endmodule

```

Chapter 2

Verification, Co-Simulation & Waveform Analysis

2.1 IP Block 5: DDS Synthesizer Verification

2.1.1 Digital Simulation (Icarus Verilog & GTKWave)

Before mixed-signal translation, the accumulator wrap-around boundaries and LUT routing were verified natively using a self-checking testbench in Icarus Verilog. The testbench dynamically altered the `tuning_word` mid-simulation from 500 to 2000 to verify instantaneous frequency hopping capabilities.



Figure 2.1: GTKWave Digital Simulation demonstrating instantaneous frequency scaling upon dynamic tuning word adjustment.

As evidenced by the digital waveform data (Figure 2.1), adjusting the frequency multiplier seamlessly tightens the accumulator sawtooth wave. The 8-bit output directly tracks this compression without introducing phase jitter or delay, proving the digital boundary logic is mathematically stable.

2.1.2 System Integration (eSim Mixed-Signal Scope)

To verify physical analog synthesis, the RTL was instantiated in the eSim NGVeri mixed-signal environment.

Schematic Wiring & ADC/DAC Boundaries

The 10 MHz master clock and reset pins were driven by analog `vpulse` sources routed through a 2-bit `adc_bridge`. To avoid unnecessary schematic clutter and efficiently simulate a constant frequency, the 16-bit `tuning_word` bus was hardwired to a static binary value of 2000 (0000 0111 1101 0000) using standard 5V `vdc` sources mapped directly to the respective ADC pins. On the output stage, the 8-bit digital sine wave bus (`sine_out[7:0]`) was routed through an 8-bit `dac_bridge` to facilitate continuous analog visualization in Ngspice.

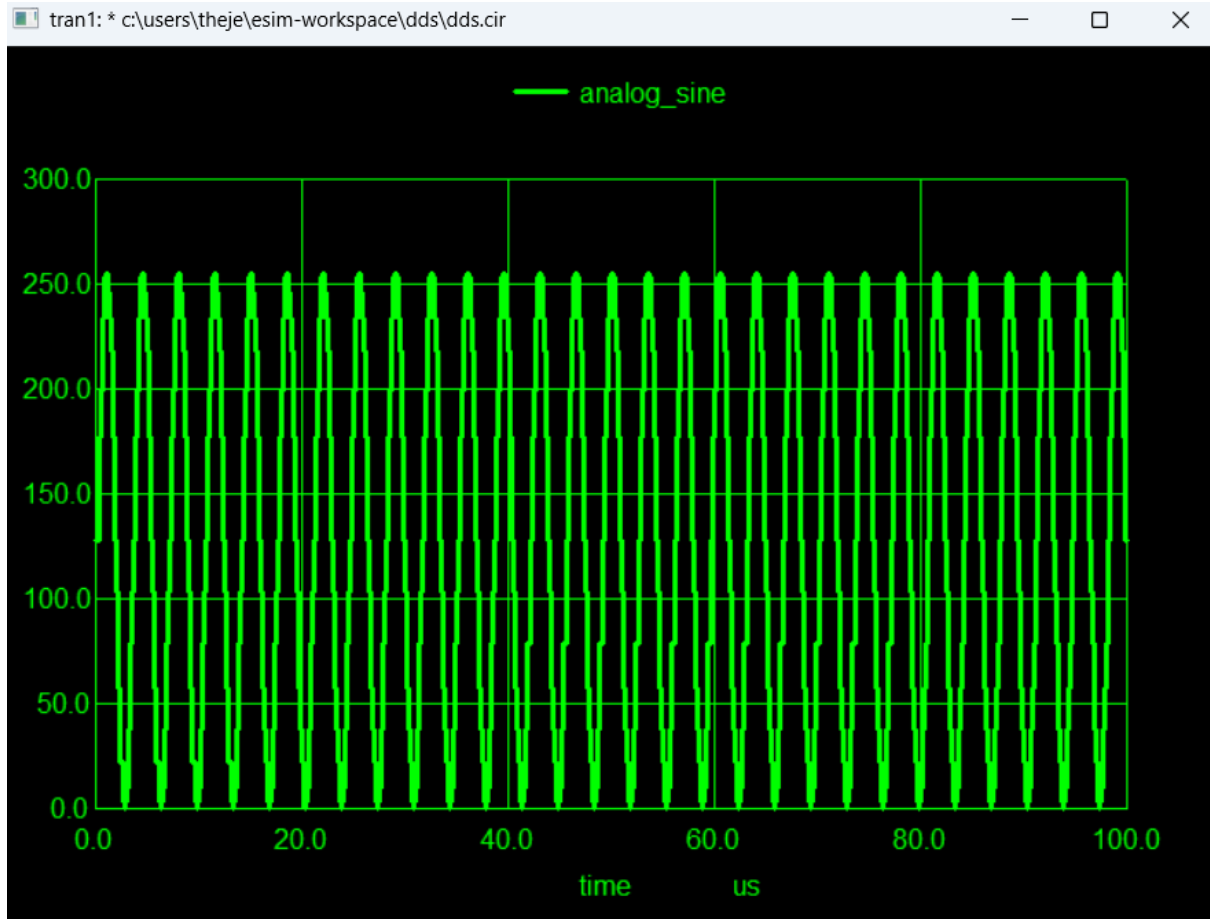


Figure 2.3: Ngspice Transient Simulation demonstrating the mathematically reconstructed 8-bit analog sine wave oscillating seamlessly between 0 and 255.

sweeps the full dynamic range of the 8-bit architecture, peaking smoothly at an amplitude of 255 and crossing cleanly through the median at 128 without combinational glitching.

2. **Frequency Stability:** Across the $100\ \mu\text{s}$ transient analysis window, the wave exhibits a perfectly uniform period. The lack of phase drift confirms that the 16-bit phase accumulator interacts flawlessly with the continuous 10 MHz eSim system clock boundary.